

NEMA IQ – Summary (VPHD, mean of 3 replicates)

GE MEDICAL SYSTEMS Omni Legend | 32 cm AFOV | SW pet_omni.42 | study 2026-09-10 | ID PHANTOM

Recon VPHD (6 it / 22 ss) | 384x384 | FOV 40 cm | filter 2 mm | pixel 1.042 mm | slice 2.07 mm | dose 84.95 MBq @ 09:18:00 | vol 9300 mL | aH/aB = 4.00

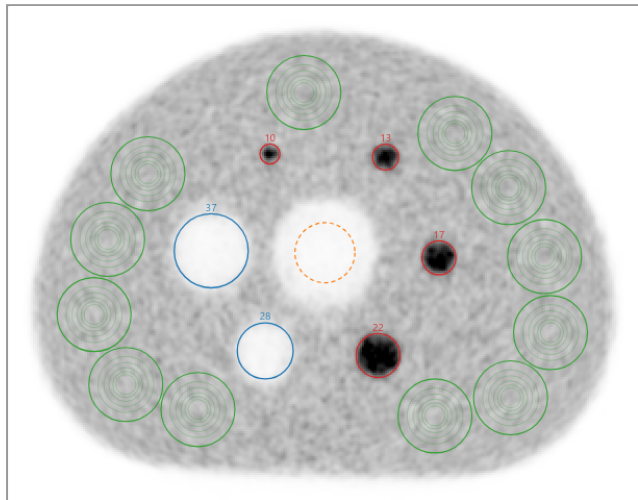
Analysis standard: NEMA NU 2-2012 · limits: No acceptance limits · report generated 2026-09-26 14:37 · NEMA IQ Analyzer v2.0.0

Overall: NO LIMITS

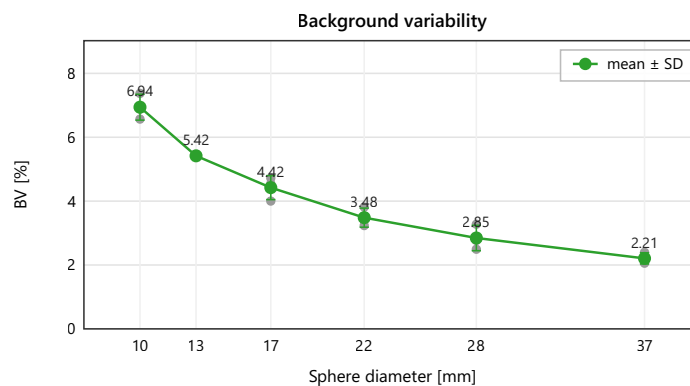
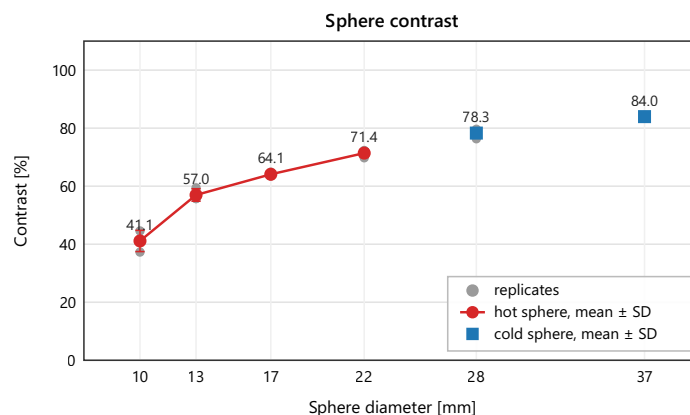
Sphere fill matches NEMA NU 2-2012 (10-22 mm hot, 28/37 mm cold).

Replicate mean, common central slice 75

■ hot ■ cold ■ background ■ lung

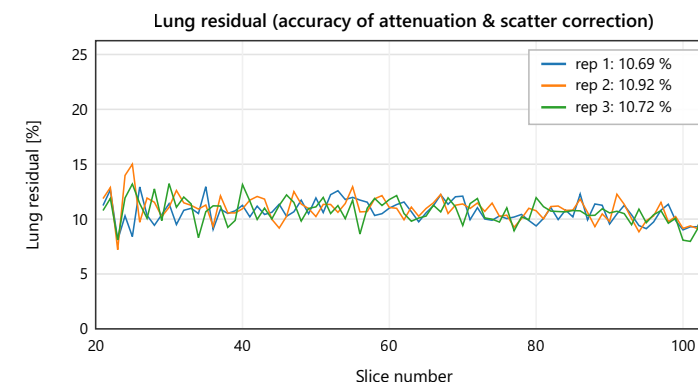


D [mm]	Q rep 1	Q rep 2	Q rep 3	BV rep 1	BV rep 2	BV rep 3
10	44.6	41.5	37.3	6.57	7.37	6.88
13	59.4	55.7	55.7	5.44	5.39	5.42
17	63.5	64.8	63.9	4.72	4.00	4.54
22	72.3	72.1	69.8	3.81	3.23	3.40
28	78.9	79.6	76.3	3.28	2.77	2.49
37	84.0	84.3	83.6	2.40	2.06	2.17
lung	10.69	10.92	10.72			
C _{B,37}	6.259	5.949	5.686	kBq/mL		



D [mm]	type	Contrast [%]	limit		BV [%]	limit	
10	hot	41.1 ± 3.7	-	N/A	6.94 ± 0.41	-	N/A
13	hot	57.0 ± 2.1	-	N/A	5.42 ± 0.03	-	N/A
17	hot	64.1 ± 0.7	-	N/A	4.42 ± 0.38	-	N/A
22	hot	71.4 ± 1.4	-	N/A	3.48 ± 0.30	-	N/A
28	cold	78.3 ± 1.7	-	N/A	2.85 ± 0.40	-	N/A
37	cold	84.0 ± 0.4	-	N/A	2.21 ± 0.17	-	N/A
Lung error		10.77 ± 0.13	-	N/A			

Contrast ≥ limit, BV ≤ limit, lung error ≤ limit; limits: No acceptance limits; contrast limits apply to hot spheres. **No acceptance limits entered: no PASS/FAIL.** N/E = limit applies but the sphere was not hot (not evaluable); N/A = no limit.



NEMA IQ – QA details

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Recon VPHD (6 it / 22 ss) | 384x384 | FOV 40 cm | filter 2 mm | pixel 1.042 mm | slice 2.07 mm | dose 84.95 MBq @ 09:18:00 | vol 9300 mL | aH/aB = 4.00
Analysis standard: NEMA NU 2-2012 · limits: No acceptance limits · report generated 2026-09-26 14:37 · NEMA IQ Analyzer v2.0.0

Protocol item (no published protocol selected)	expected	found	status
Scan time replicate 1 [s]	-	425	INFO
Scan time replicate 2 [s]	-	444	INFO
Scan time replicate 3 [s]	-	465	INFO
Number of replicates	-	3	INFO
Implied bed step (replicate 1), NEMA 100 cm / 30 min [cm]	-	23.6 (= 26 % bed overlap for 31.7 cm axial FOV)	INFO
Recon method	-	VPHD	INFO
Image size	-	384x384	INFO
Pixel size [mm]	-	1.042	INFO
Recon FOV [cm]	-	40	INFO
Iterations	-	6	INFO
Subsets	-	22	INFO
Post-filter FWHM [mm]	-	2	INFO
z-filter	-	none	INFO
Attenuation corrected	ATTN	yes	OK
Scatter corrected	SCAT	yes	OK
Units	BQML	BQML	OK
Decay correction	START / ADMIN	START	OK

Acquisition

rep 1: NEMA IQ_Frame1 — start 10:10:33, 425 s, prompts 958 M, delays 441 M
rep 2: NEMA IQ_Frame2 — start 10:18:05, 444 s, prompts 937 M, delays 420 M
rep 3: NEMA IQ_Frame3 — start 10:25:41, 465 s, prompts 919 M, delays 400 M

Activity concentration (background, 37 mm ROIs)

rep	C _{B,37} measured	expected	meas/exp
1	6.259	6.534	0.958
2	5.949	6.231	0.955
3	5.686	5.939	0.957

kBq/mL, each replicate decay-corrected to its own start. Expected = net dose 84.95 MBq at 2026-09-10 09:18:00 / volume 9300 mL (pre 94.72 MBq, residual 9.18 MBq). NEMA NU 2 recommended background at the start of the test: 5.3 kBq/mL (spheres 21.2 kBq/mL at a_H/a_B = 4.00). Replicate consistency after decay correction: CV 0.17 %.

Measured sphere/background ratio

D	type	C _S /C _B rep 1	C _S /C _B rep 2	C _S /C _B rep 3
10	hot	2.34	2.25	2.12
13	hot	2.78	2.67	2.67
17	hot	2.91	2.95	2.92
22	hot	3.17	3.16	3.09
28	cold	0.21	0.20	0.24
37	cold	0.16	0.16	0.16

CTAC check

CT slice z	26.3 mm (140 kVp)
Water (12 BG ROIs)	0.4 HU [-0.9, 2.7], pixel SD 3.6
Lung insert (30 mm)	-695 HU
Sphere contents	10: -2, 13: 5, 17: 2, 22: 0, 28: -0, 37: 0 HU

Overall: NO LIMITS

Geometry (automatic localisation)

Common central slice	75 (z = 26.3 mm)
Sphere centres (slice)	10: 75.4, 13: 75.3, 17: 75.0, 22: 75.1, 28: 76.3, 37: 76.2
Sphere ROI slices	10: 75, 13: 75, 17: 75, 22: 75, 28: 76, 37: 76 (per-sphere mode)
Sphere ring	radius 57.1 mm (57.2), rms residual 0.36 mm
Tilt / axial spread	lung axis 1.10°, sphere centres 2.7 mm
Background ROIs	margin 11.5 mm (NEMA 15), wall gap 11.5–11.5 mm, sphere gap ≥ 11.6 mm, spacing 38.1 mm
Background slices	65, 70, 75, 80, 85 (-20.7, -10.3, +0.0, +10.3, +20.7 mm)
Lung slices	21–103 (n = 83); insert 15.5–108.5; exclusion 10 mm

Standard

	NEMA NU 2-2012	NEMA NU 2-2018
Hot spheres	10, 13, 17, 22 mm	all six (10–37 mm)
Cold spheres	28, 37 mm (Q _C)	none
Lung slices excluded	10 mm from insert ends	30 mm from insert ends
Used here	✓	

Warnings / notes

- sphere centres span 2.7 mm axially (phantom tilt 1.1 deg); each sphere ROI is placed on its own central slice
- NEMA 15 mm wall/sphere margins cannot be met by 12 non-overlapping 37 mm background ROIs in this phantom configuration; margin reduced to 11.5 mm (min ROI spacing 38.1 mm)

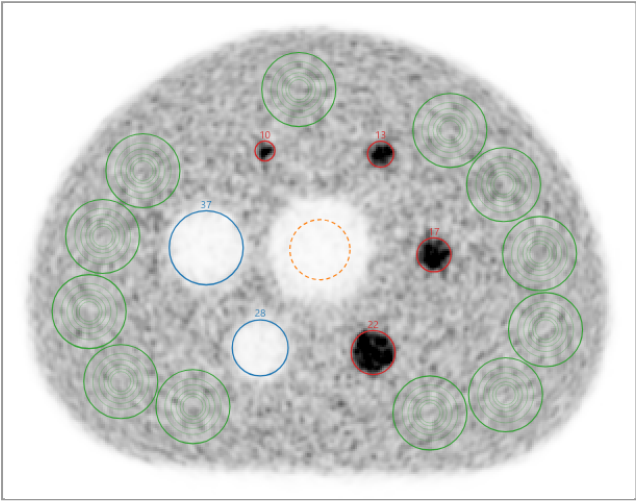
Definitions (NEMA NU 2, Sec. 7)

Q_H = (C_H/C_B - 1)/(a_H/a_B - 1) × 100 %; Q_C = (1 - C_C/C_B) × 100 %; BV = SD_B/C_B × 100 % over 60 background ROIs (12 ROIs × 5 slices, K-1); lung error = C_{lung,i}/C_{B,37} × 100 %, averaged over slices. ROI means use fractional (sub-pixel) weights. a_H/a_B = 4.000.

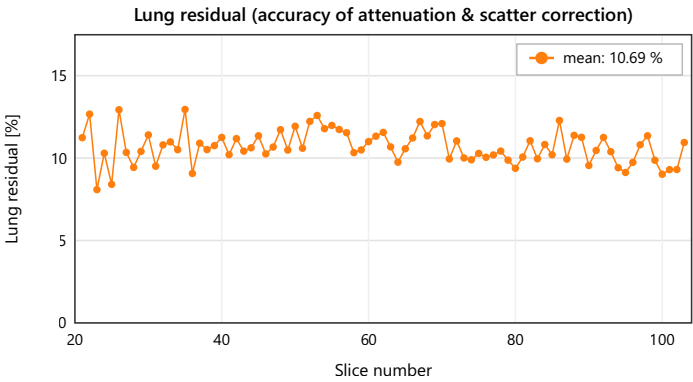
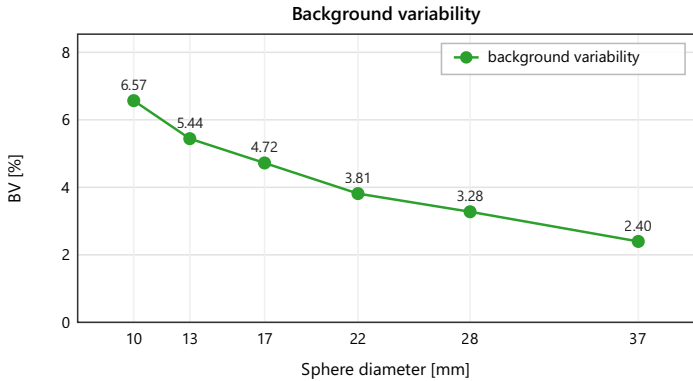
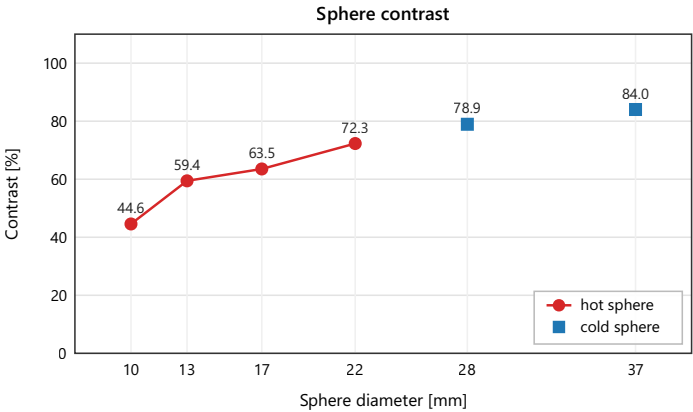
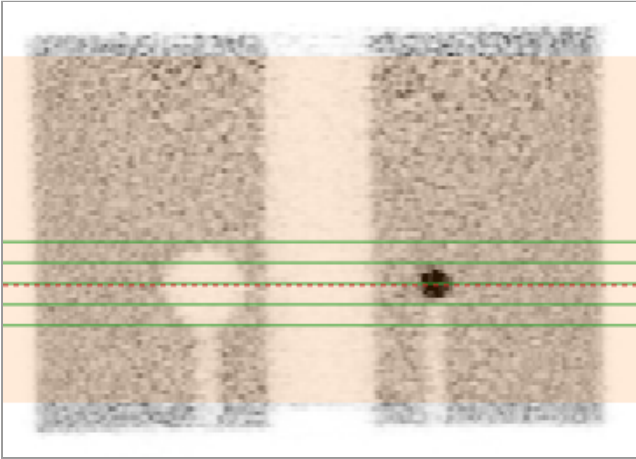
NEMA IQ – Replicate 1/3: NEMA IQ_Frame1 (start 10:10:33, 425 s)

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Common central slice 75; sphere ROI slices 10:75, 13:75, 17:75, 22:75, 28:76, 37:76



Coronal through lung axis — green: background slices, orange: lung range, red: sphere plane

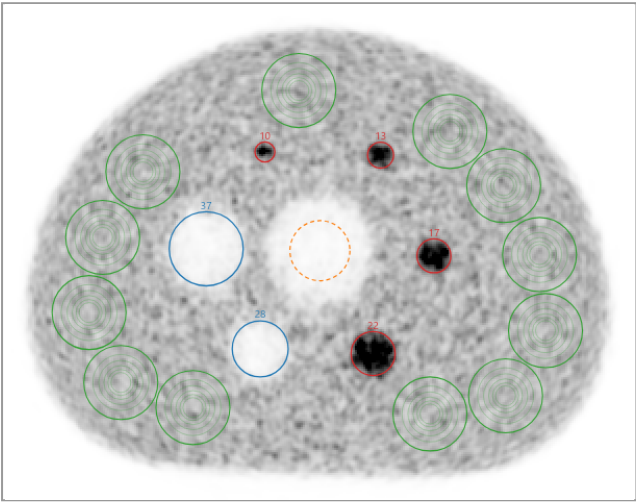


D	type	slice	Q [%]	BV [%]	C _S [kBq/mL]	C _B [kBq/mL]
10	hot	75	44.6	6.57	14.649	6.267
13	hot	75	59.4	5.44	17.379	6.244
17	hot	75	63.5	4.72	18.187	6.258
22	hot	75	72.3	3.81	19.856	6.265
28	cold	76	78.9	3.28	1.318	6.260
37	cold	76	84.0	2.40	0.999	6.259
Lung error			10.69 % (max 12.95 %)			
C _{B,37}			6.259 kBq/mL (expected 6.534, ratio 0.958)			

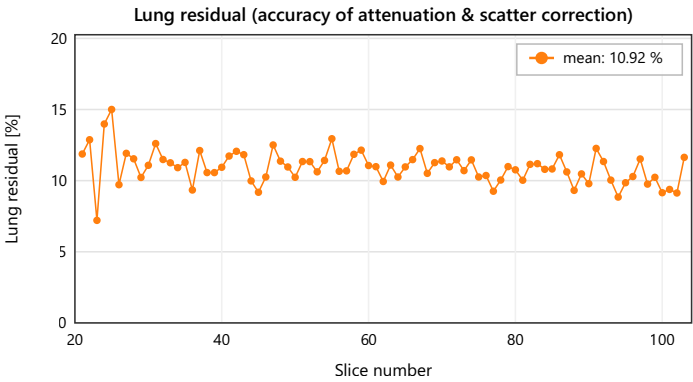
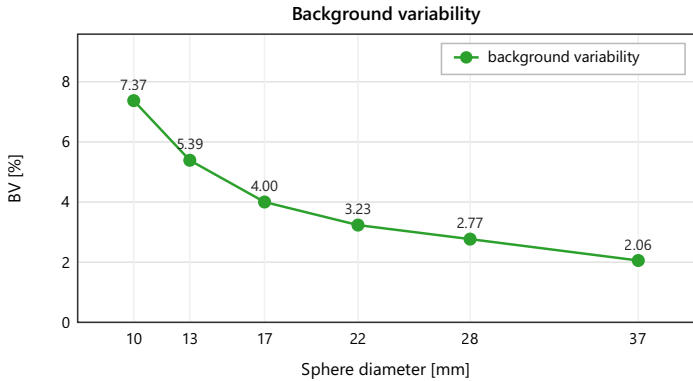
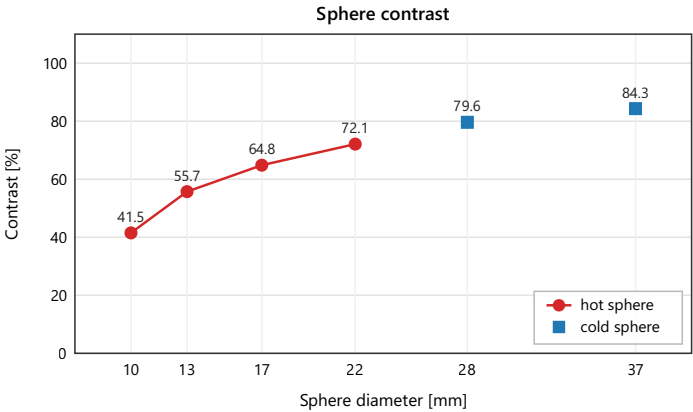
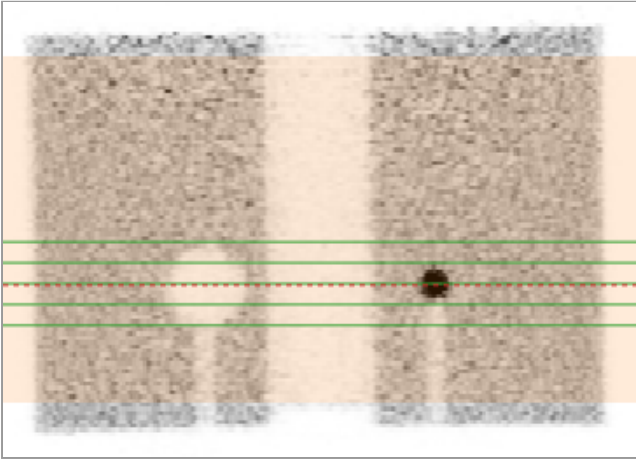
NEMA IQ – Replicate 2/3: NEMA IQ_Frame2 (start 10:18:05, 444 s)

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Analysis standard: NEMA NU 2-2012 · limits: No acceptance limits · report generated 2026-09-26 14:37 · NEMA IQ Analyzer v2.0.0

Common central slice 75; sphere ROI slices 10:75, 13:75, 17:75, 22:75, 28:76, 37:76



Coronal through lung axis — green: background slices, orange: lung range, red: sphere plane

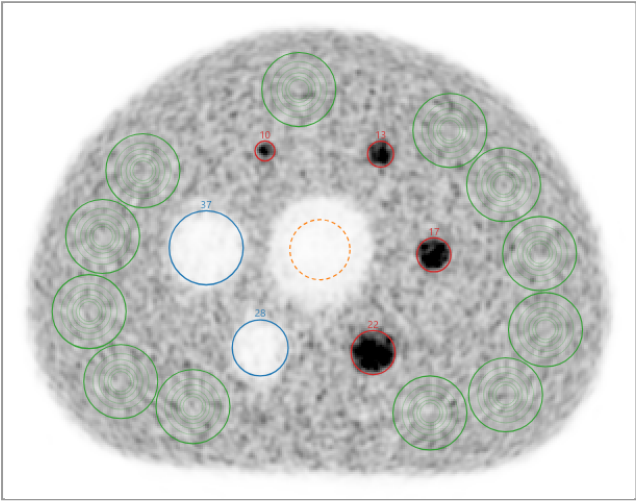


D	type	slice	Q [%]	BV [%]	C _S [kBq/mL]	C _B [kBq/mL]
10	hot	75	41.5	7.37	13.350	5.946
13	hot	75	55.7	5.39	15.858	5.936
17	hot	75	64.8	4.00	17.503	5.943
22	hot	75	72.1	3.23	18.859	5.962
28	cold	76	79.6	2.77	1.212	5.946
37	cold	76	84.3	2.06	0.932	5.949
Lung error			10.92 % (max 15.00 %)			
C _{B,37}			5.949 kBq/mL (expected 6.231, ratio 0.955)			

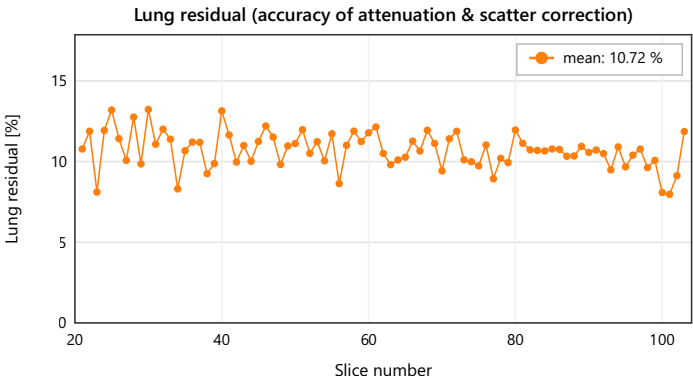
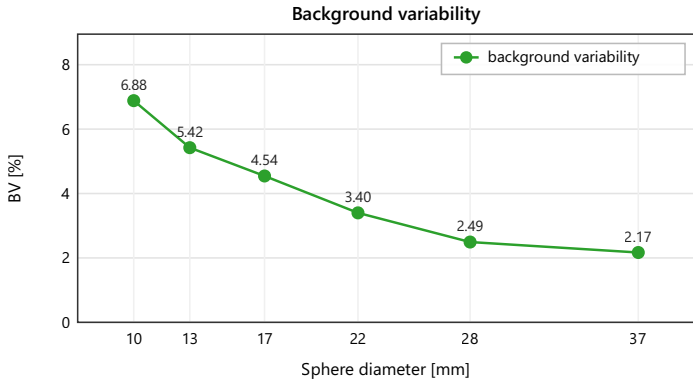
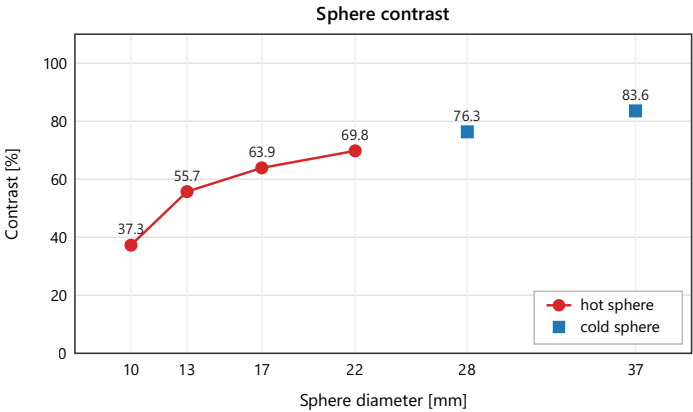
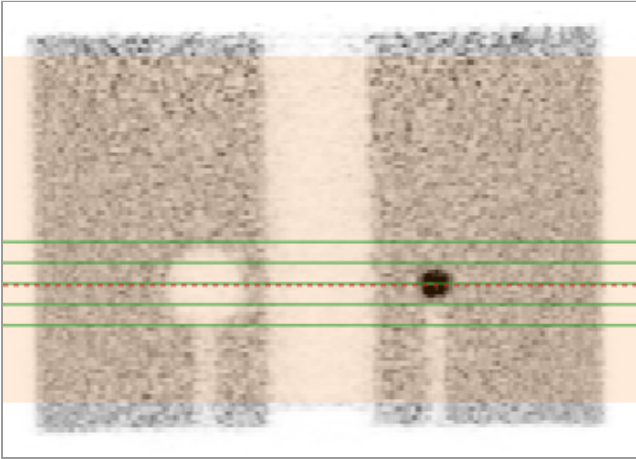
NEMA IQ – Replicate 3/3: NEMA IQ_Frame3 (start 10:25:41, 465 s)

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Common central slice 75; sphere ROI slices 10:75, 13:75, 17:75, 22:75, 28:76, 37:76



Coronal through lung axis — green: background slices, orange: lung range, red: sphere plane



D	type	slice	Q [%]	BV [%]	C _S [kBq/mL]	C _B [kBq/mL]
10	hot	75	37.3	6.88	11.861	5.598
13	hot	75	55.7	5.42	15.011	5.618
17	hot	75	63.9	4.54	16.450	5.639
22	hot	75	69.8	3.40	17.515	5.662
28	cold	76	76.3	2.49	1.344	5.679
37	cold	76	83.6	2.17	0.935	5.686
Lung error			10.72 % (max 13.23 %)			
C _{B,37}			5.686 kBq/mL (expected 5.939, ratio 0.957)			

NEMA IQ – Appendix: ROI data and lung residual per slice

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Sphere ROIs (ROI diameter = sphere inner diameter)

D	type	slice	x, y [px]	C _S rep 1	C _S rep 2	C _S rep 3	C _B rep 1	C _B rep 2	C _B rep 3
10	hot	75	165.3, 140.5	14.649	13.350	11.861	6.267	5.946	5.598
13	hot	75	220.8, 142.0	17.379	15.858	15.011	6.244	5.936	5.618
17	hot	75	246.4, 190.3	18.187	17.503	16.450	6.258	5.943	5.639
22	hot	75	217.2, 237.2	19.856	18.859	17.515	6.265	5.962	5.662
28	cold	76	163.0, 235.0	1.318	1.212	1.344	6.260	5.946	5.679
37	cold	76	137.1, 187.0	0.999	0.932	0.935	6.259	5.949	5.686

Concentrations in kBq/mL. C_B = mean of the 60 background ROIs of the sphere's diameter.

Background ROI centres

#	x, y [px]	wall gap [mm]	sphere gap [mm]
1	254.1, 130.7	11.5	11.6
2	279.9, 156.6	11.5	22.5
3	297.1, 189.7	11.5	25.8
4	300.0, 226.2	11.5	40.2
5	280.8, 257.5	11.5	40.0
6	244.5, 266.2	11.5	12.0
7	130.7, 263.2	11.5	12.2
8	96.1, 251.1	11.5	39.2
9	80.9, 217.6	11.5	29.7
10	87.4, 181.5	11.5	15.1
11	106.7, 150.0	11.5	12.9
12	181.6, 111.0	11.5	11.6

Background slices 65, 70, 75, 80, 85.

Lung residual per slice [%] (30 mm ROI / C_{B,37})

slice	z [mm]	rep 1	rep 2	rep 3
21	-85.5	11.24	11.87	10.78
22	-83.5	12.67	12.88	11.88
23	-81.4	8.09	7.21	8.11
24	-79.3	10.29	13.98	11.93
25	-77.3	8.41	15.00	13.19
26	-75.2	12.93	9.71	11.42
27	-73.1	10.34	11.92	10.07
28	-71.0	9.44	11.53	12.76
29	-69.0	10.40	10.22	9.85
30	-66.9	11.41	11.08	13.23
31	-64.8	9.51	12.61	11.08
32	-62.8	10.80	11.49	12.01
33	-60.7	10.98	11.25	11.39
34	-58.6	10.51	10.91	8.31
35	-56.5	12.95	11.28	10.68
36	-54.5	9.07	9.34	11.21
37	-52.4	10.90	12.12	11.19
38	-50.3	10.52	10.57	9.25
39	-48.3	10.76	10.57	9.88
40	-46.2	11.26	10.94	13.14
41	-44.1	10.21	11.73	11.65
42	-42.1	11.18	12.06	9.97
43	-40.0	10.42	11.82	11.00
44	-37.9	10.62	9.98	10.02
45	-35.9	11.35	9.18	11.25
46	-33.8	10.25	10.26	12.21
47	-31.7	10.67	12.51	11.52
48	-29.6	11.72	11.37	9.82
49	-27.6	10.49	10.96	10.97
50	-25.5	11.93	10.24	11.12
51	-23.4	10.60	11.34	11.97
52	-21.4	12.23	11.34	10.51
53	-19.3	12.59	10.62	11.23
54	-17.2	11.78	11.42	10.04
55	-15.2	11.98	12.95	11.72
56	-13.1	11.73	10.66	8.64
57	-11.0	11.54	10.69	11.01
58	-8.9	10.33	11.85	11.89
59	-6.9	10.50	12.14	11.24
60	-4.8	11.00	11.06	11.78
61	-2.7	11.33	10.98	12.15
62	-0.7	11.56	9.94	10.51

slice	z [mm]	rep 1	rep 2	rep 3
63	1.4	10.68	11.09	9.81
64	3.5	9.75	10.25	10.10
65	5.5	10.57	10.97	10.27
66	7.6	11.22	11.48	11.27
67	9.7	12.22	12.25	10.66
68	11.8	11.35	10.51	11.94
69	13.8	12.03	11.27	11.12
70	15.9	12.09	11.38	9.42
71	18.0	9.95	10.97	11.41
72	20.0	11.04	11.47	11.88
73	22.1	10.01	10.70	10.11
74	24.2	9.90	11.46	9.99
75	26.3	10.28	10.26	9.73
76	28.3	10.05	10.36	11.03
77	30.4	10.19	9.26	8.93
78	32.5	10.42	10.04	10.20
79	34.5	9.88	10.99	9.94
80	36.6	9.38	10.76	11.95
81	38.7	10.06	10.03	11.13
82	40.7	11.05	11.14	10.73
83	42.8	9.96	11.19	10.69
84	44.9	10.82	10.80	10.66
85	47.0	10.20	10.82	10.78
86	49.0	12.28	11.82	10.75
87	51.1	9.94	10.61	10.33
88	53.2	11.38	9.32	10.35
89	55.2	11.26	10.47	10.94
90	57.3	9.55	9.79	10.56
91	59.4	10.47	12.27	10.72
92	61.4	11.25	11.34	10.50
93	63.5	10.39	10.04	9.49
94	65.6	9.41	8.84	10.91
95	67.7	9.13	9.85	9.67
96	69.7	9.74	10.29	10.40
97	71.8	10.81	11.52	10.77
98	73.9	11.35	9.76	9.63
99	75.9	9.87	10.23	10.07
100	78.0	9.02	9.15	8.08
101	80.1	9.30	9.38	7.97
102	82.1	9.31	9.13	9.13
103	84.2	10.95	11.64	11.86